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HW5 - CS5402: Introduction to Data Mining

Task 1: Self-Attention (Theory):

Consider an input sequence:

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} (n = 3,\ d_{model} = 2)$$

Let:

$$W_q = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, W_k = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, W_v = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Q1: Compute Q, K, V and show all intermediate steps.
Equation: $Q = XW_q$, $K = XW_k$, $V = XW_v$
(Working out problem in LaTeX to show matrices):

$$Q = XW_q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (1\cdot 1 + 0\cdot 0) & (1\cdot 1 + 0\cdot 1) \\ (0\cdot 1 + 1\cdot 0) & (0\cdot 1 + 1\cdot 1) \\ (1\cdot 1 + 1\cdot 0) & (1\cdot 1 + 1\cdot 1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix}$$

$$K = XW_k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} (1\cdot 1 + 0\cdot 1) & (1\cdot 0 + 0\cdot 1) \\ (0\cdot 1 + 1\cdot 1) & (0\cdot 0 + 1\cdot 1) \\ (1\cdot 1 + 1\cdot 1) & (1\cdot 0 + 1\cdot 1) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$V = XW_v = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} (1\cdot 1 + 0\cdot 0) & (1\cdot 0 + 0\cdot 2) \\ (0\cdot 1 + 1\cdot 0) & (0\cdot 0 + 1\cdot 2) \\ (1\cdot 1 + 1\cdot 0) & (1\cdot 0 + 1\cdot 2) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

Q2: Compute Attention Scores
Equation: $S = QK^T$
(Working out problem in LaTeX to show matrices):

$$S = QK^T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} (1\cdot 1 + 1\cdot 0) & (1\cdot 1 + 1\cdot 1) & (1\cdot 2 + 1\cdot 1) \\ (0\cdot 1 + 1\cdot 0) & (0\cdot 1 + 1\cdot 1) & (0\cdot 2 + 1\cdot 1) \\ (1\cdot 1 + 2\cdot 0) & (1\cdot 1 + 2\cdot 1) & (1\cdot 2 + 2\cdot 1) \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix}$$

Q3: Apply Scaling and SoftMax (Apply row-wise softmax, and Keep at least 3 decimal places)

Equation: $\hat{S} = \frac{S}{\sqrt{d_k}}$

Since K is 2-dimensional (2 columns), $d_k = 2$

(Working out problem in LaTeX to show matrices):

Scaling:

$$\hat{S} = \frac{S}{\sqrt{d_k}} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{2}} & \frac{3}{\sqrt{2}} \\ \frac{0}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{3}{\sqrt{2}} & \frac{4}{\sqrt{2}} \end{bmatrix} \approx \begin{bmatrix} 0.707 & 1.414 & 2.121 \\ 0.000 & 0.707 & 0.707 \\ 0.707 & 2.121 & 2.828 \end{bmatrix}$$

Softmax:

$$\text{Softmax}(\hat{S}) = \begin{bmatrix} \frac{e^{0.707}}{e^{0.707}+e^{1.414}+e^{2.121}} & \frac{e^{1.414}}{e^{0.707}+e^{1.414}+e^{2.121}} & \frac{e^{2.121}}{e^{0.707}+e^{1.414}+e^{2.121}} \\ \frac{e^{0.000}}{e^{0.000}+e^{0.707}+e^{0.707}} & \frac{e^{0.707}}{e^{0.000}+e^{0.707}+e^{0.707}} & \frac{e^{0.707}}{e^{0.000}+e^{0.707}+e^{0.707}} \\ \frac{e^{0.707}}{e^{0.707}+e^{2.121}+e^{2.828}} & \frac{e^{2.121}}{e^{0.707}+e^{2.121}+e^{2.828}} & \frac{e^{2.828}}{e^{0.707}+e^{2.121}+e^{2.828}} \end{bmatrix} \approx \begin{bmatrix} \frac{2.028}{14.479} & \frac{4.112}{14.479} & \frac{8.339}{14.479} \\ \frac{1.000}{5.056} & \frac{2.028}{5.056} & \frac{2.028}{5.056} \\ \frac{2.028}{27.279} & \frac{8.339}{27.279} & \frac{16.912}{27.279} \end{bmatrix} \approx \begin{bmatrix} 0.140 & 0.284 & 0.576 \\ 0.198 & 0.401 & 0.401 \\ 0.074 & 0.306 & 0.620 \end{bmatrix}$$

$$\text{Softmax}(\hat{S}) = \begin{bmatrix} \frac{e^{0.707}}{e^{0.707}+e^{1.414}+e^{2.121}} & \frac{e^{1.414}}{e^{0.707}+e^{1.414}+e^{2.121}} & \frac{e^{2.121}}{e^{0.707}+e^{1.414}+e^{2.121}} \\ \frac{e^{0.000}}{e^{0.000}+e^{0.707}+e^{0.707}} & \frac{e^{0.707}}{e^{0.000}+e^{0.707}+e^{0.707}} & \frac{e^{0.707}}{e^{0.000}+e^{0.707}+e^{0.707}} \\ \frac{e^{0.707}}{e^{0.707}+e^{2.121}+e^{2.828}} & \frac{e^{2.121}}{e^{0.707}+e^{2.121}+e^{2.828}} & \frac{e^{2.828}}{e^{0.707}+e^{2.121}+e^{2.828}} \end{bmatrix}$$

$$\approx \begin{bmatrix} \frac{2.028}{14.479} & \frac{4.112}{14.479} & \frac{8.339}{14.479} \\ \frac{1.000}{5.056} & \frac{2.028}{5.056} & \frac{2.028}{5.056} \\ \frac{2.028}{27.279} & \frac{8.339}{27.279} & \frac{16.912}{27.279} \end{bmatrix} \approx \begin{bmatrix} 0.140 & 0.284 & 0.576 \\ 0.198 & 0.401 & 0.401 \\ 0.074 & 0.306 & 0.620 \end{bmatrix}$$

Q4: Compute Final Output Attention(Q,K, V)

$Attention(Q, K, V) = Softmax(\widehat{S}) \times V$

$$\text{Attention}(Q, K, V) = \text{Softmax}(\hat{S}) \cdot V = \begin{bmatrix} 0.140 & 0.284 & 0.576 \\ 0.198 & 0.401 & 0.401 \\ 0.074 & 0.306 & 0.620 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} (0.140 \cdot 1 + 0.284 \cdot 0 + 0.576 \cdot 1) & (0.140 \cdot 0 + 0.284 \cdot 2 + 0.576 \cdot 2) \\ (0.198 \cdot 1 + 0.401 \cdot 0 + 0.401 \cdot 1) & (0.198 \cdot 0 + 0.401 \cdot 2 + 0.401 \cdot 2) \\ (0.074 \cdot 1 + 0.306 \cdot 0 + 0.620 \cdot 1) & (0.074 \cdot 0 + 0.306 \cdot 2 + 0.620 \cdot 2) \end{bmatrix}$$

$$= \begin{bmatrix} (0.140 + 0 + 0.576) & (0 + 0.568 + 1.152) \\ (0.198 + 0 + 0.401) & (0 + 0.802 + 0.802) \\ (0.074 + 0 + 0.620) & (0 + 0.612 + 1.240) \end{bmatrix} = \begin{bmatrix} 0.716 & 1.720 \\ 0.599 & 1.604 \\ 0.694 & 1.852 \end{bmatrix}$$

Final Output:

$$\text{Attention}(Q, K, V) = \begin{bmatrix} 0.716 & 1.720 \\ 0.599 & 1.604 \\ 0.694 & 1.852 \end{bmatrix}$$

Task 2: Multi-Head Attention Implementation:

Experiment 1: Effect of Scaling

- Run attention with scaling:

```
=== With Scaling: ===
after split: Q shape: (2, 2, 4, 4)
after split: K shape: (2, 2, 4, 4)
after split: V shape: (2, 2, 4, 4)
attention matrix (head 0):
 [[5.48696357e-03 8.78476429e-09 7.19869820e-09 9.94513020e-01]
 [1.70457118e-02 2.60834934e-05 1.41732657e-06 9.82926787e-01]
 [1.16682417e-01 4.30958397e-01 4.52355246e-01 3.93994633e-06]
 [4.10776724e-01 1.22482145e-07 1.73702554e-08 5.89223136e-01]]
Output shape: (2, 4, 8)
```

- Run attention without scaling:

```
=== Without Scaling: ===
after split: Q shape: (2, 2, 4, 4)
after split: K shape: (2, 2, 4, 4)
after split: V shape: (2, 2, 4, 4)
attention matrix (head 0):
 [[3.04389724e-05 7.80236135e-17 5.23930603e-17 9.99969561e-01]
 [3.00647331e-04 7.03977180e-10 2.07858086e-12 9.99699352e-01]
 [3.37028701e-02 4.59755301e-01 5.06541829e-01 3.84270162e-11]
 [3.27060498e-01 2.90778311e-14 5.84828932e-16 6.72939502e-01]]
Output shape: (2, 4, 8)
```

Experiment 2: Effect of Multi-Head

- Run your model with Multi-head (e.g., num_heads = 2 or 4)

```
=== Multi-Head (4 Heads): ===
after split: Q shape: (2, 4, 4, 2)
after split: K shape: (2, 4, 4, 2)
after split: V shape: (2, 4, 4, 2)
attention matrix (head 0):
 [[2.91406927e-01 2.02621064e-06 8.17585526e-13 7.08591047e-01]
 [4.58551858e-01 3.73506493e-04 3.23921501e-08 5.41074603e-01]
 [1.34913826e-02 7.46623750e-01 2.39733461e-01 1.51406313e-04]
 [2.62535004e-01 3.62546430e-07 2.04040745e-14 7.37464634e-01]]
Output shape: (2, 4, 8)
```

- Modify your implementation to simulate Single-head attention (num_heads = 1)

```
=== Single Head: ===
after split: Q shape: (2, 1, 4, 8)
after split: K shape: (2, 1, 4, 8)
after split: V shape: (2, 1, 4, 8)
attention matrix (head 0):
 [[6.26388684e-05 4.52030860e-07 1.77900773e-03 9.98157901e-01]
 [1.95780384e-04 1.26844711e-04 2.19189759e-02 9.77758399e-01]
 [1.78994051e-01 6.38872859e-01 1.82128525e-01 4.56467545e-06]
 [5.58473683e-07 3.09809054e-07 9.86672804e-01 1.33263276e-02]]
Output shape: (2, 4, 8)
```

Based on your experiments, answer:

1) Why is scaling needed?

- What differences do you observe with vs. without scaling?
  Without scaling, the attention weights are more extreme, focusing on a single token with values close to 0 or 1. With scaling, the weights are less extreme and multiple tokens share attention more evenly.
- How does it affect the attention distribution?
  Scaling produces a smoother attention distribution, while removing scaling leads to a sharper, more peaked distribution. Without scaling, the model over-focuses on a few tokens instead of considering all of them.

2) What happens if we remove multi-head?

Compare single-head vs multi-head outputs

- What differences do you observe?
  In the multi-head setup, each head gives a different attention matrix, showing that each head focuses on different token relationships. On the other hand, the single-head model has only one attention matrix, meaning only one representation can be tracked.
- What might be the limitation of single-head attention?
  Single-head attention has lower representation power because it can only learn one type of relationship between tokens at a time. This makes it harder to capture multiple unique patterns in the input.